Graphs

Undirected: $\{u, v\}$

Directed: (u, v)

Graph Types

Simple

- Edges undirected and unweighted
- No multi-edges (no same edge)
- No self loops (no edge starts/ends on same vertex)

Tree

Every pair of vertices only connected by exactly one path

Directed Acyclic Graph (DAG)

Directed but with no cycles

Bipartite Graph

- Two disjoint sets of vertices U and V
- For all edges $\{u, v\}$, u and v are in different vertex sets

K-Colorable

Assignment of k colors to vertices so that no pair of adjacent vertices have same color

Requires **BFS** algorithm

Uses: scheduling, make sure no two exams overlap

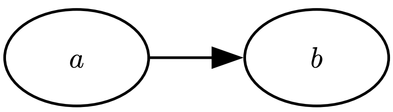
Sparse Graph

- A sparse (connected) graph has about at most as many edges as vertices
- For connected: $n - 1$ edges minimal but also could be disconnected

Dense Graph

- Has nearly the max number of edges
- Max num of edges: $n \cdot (n - 1)/2$

Graph Representations



Adjacency Matrix

	a	b
a	1	0
b	0	0

Adjacency List

a	b
b	

Types of Edges

- Tree edge: in BFS or DFS search tree
- Back edge
- Forward edge
- Cross edge: all other edges

Runtimes

BFS and DFS: $\mathrm{O}(|V| + |E|)$

Dijkstra: $\mathrm{O}((|V| + |E|) \cdot \log(|V|)) = \mathrm{O}(|E| \cdot \log(|V|))$
- Assuming a connected graph, making $|V| = \mathrm{O}(|E|)$

Dijkstra

- Uses PQ, pop non-infinity smallest, update neighbors
- Negative edge weights may cause cycles, interferes w/alg